Performed by: Vipul Jadhav

Assignment 9

Task (a): Build a WAR file using Jenkins on the master node and deploy it to a Tomcat server running on a slave instance.

Steps:

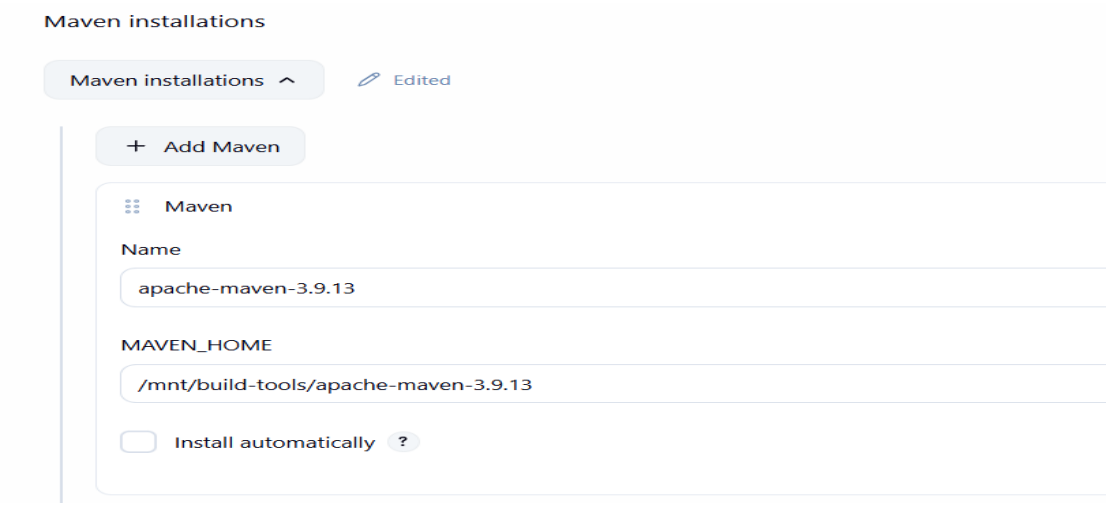
1. Launched 2 ec2 instances

slave 1	i-08fe2a658980f26cb	Running	t3.micro	3/3 checks passed	View alarms +	ap-south-1a
jenkins-master	i-0833c0ec42d2a6a8d	Running	t3.micro	3/3 checks passed	View alarms +	ap-south-1a

2. Installed java, tomcat v10, Jenkins and maven in Jenkins master manually
3. Created environment variable for maven

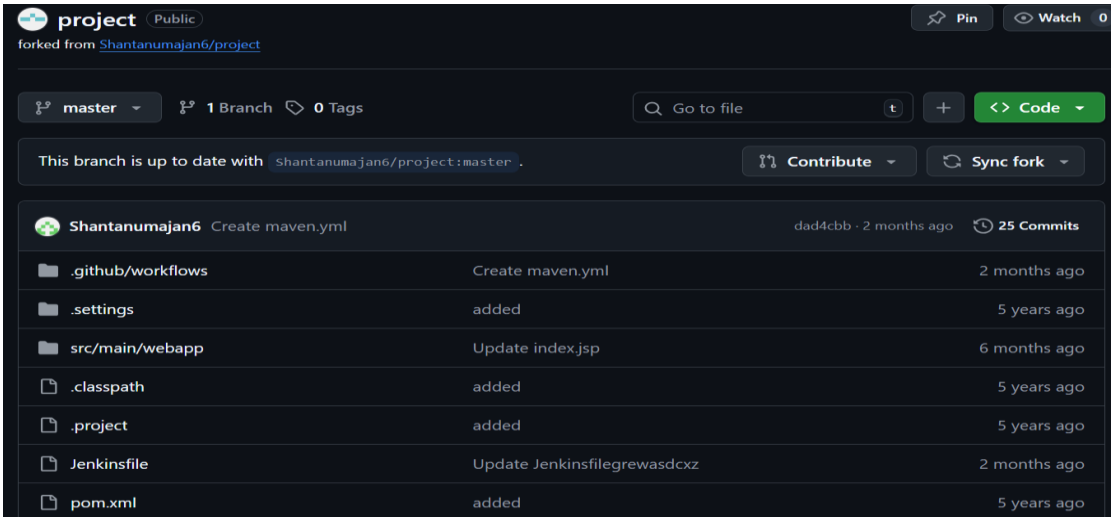
```
# .bash_profile

# Get the aliases and functions
if [ -f ~/.bashrc ]; then
        . ~/.bashrc
fi

# User specific environment and startup programs
#
#
export MAVEN_HOME=/mnt/build-tools/apache-maven-3.9.13
export PATH=$MAVEN_HOME/bin:$PATH
```

4. In Jenkins tools section uncheck install maven automatically checkbox and add name of maven (added apache-maven-3.9.13) and add the location of maven home.



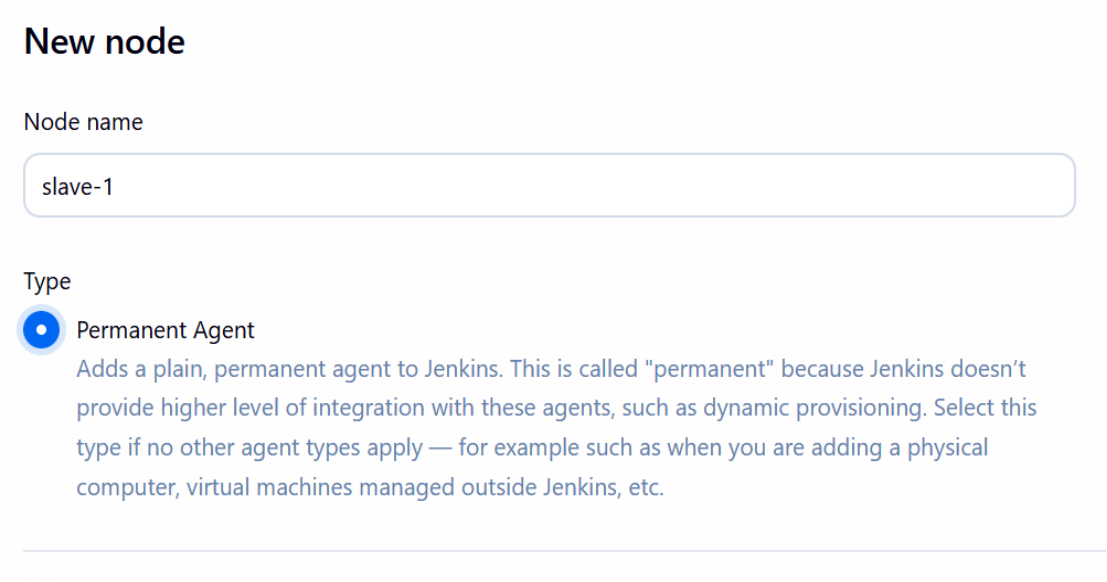
5. Created Jenkins workspace directory (/mnt/project) in master instance

6. Github repository that we are going to use.

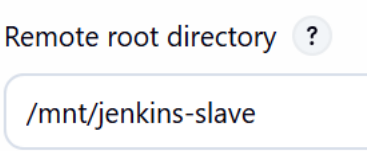


7. Copied key pair manually in Jenkins master (in /mnt/other directory) to access other ec2 instance through Jenkins.

8. Creating slave-1 node and checking permanent agent option

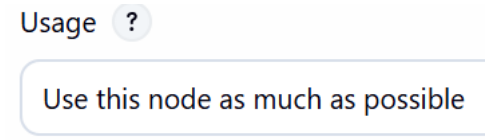


9. Created remote root directory in slave 1 and selected same remote root directory in "Remote root directory" section in slave configuration section

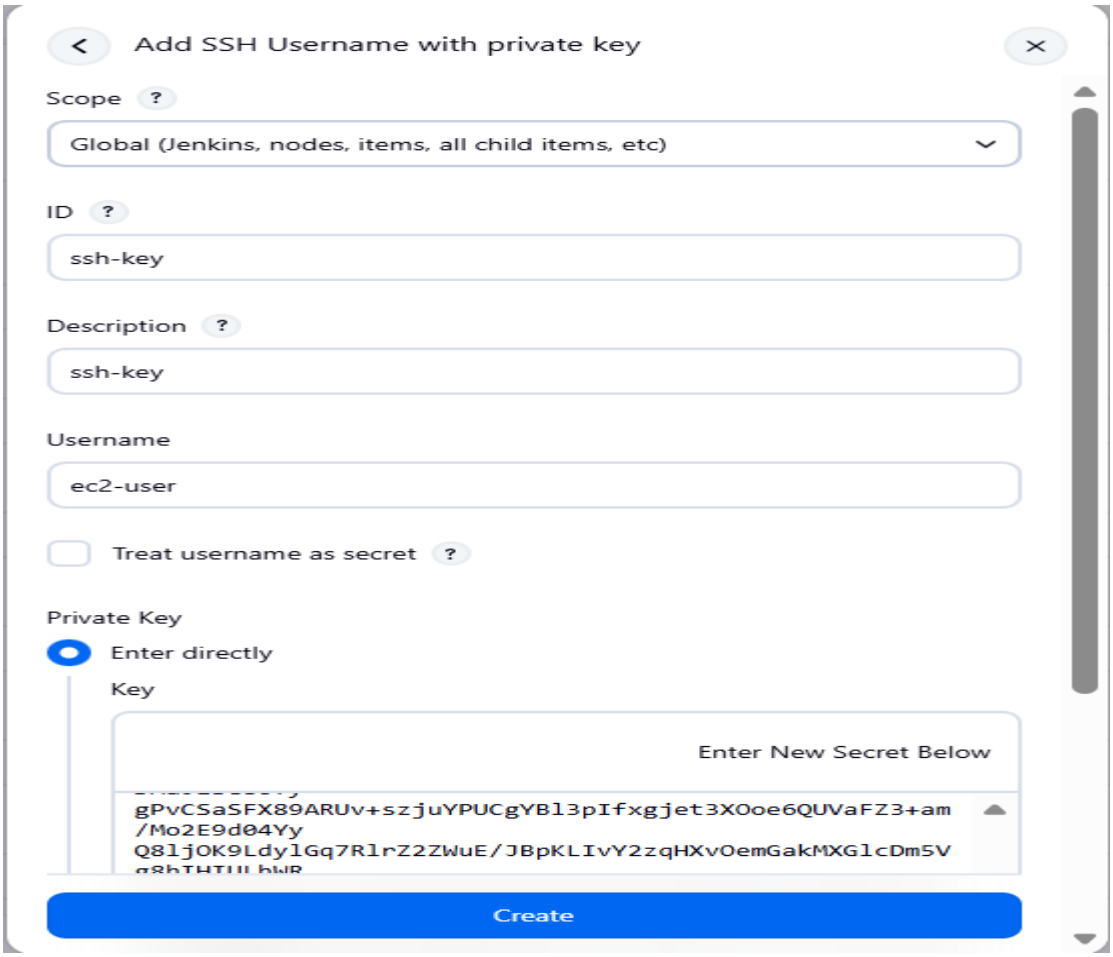


10. selected label slave 1 for slave 1 node

11. Selected usage as “Use this node as much as possible”

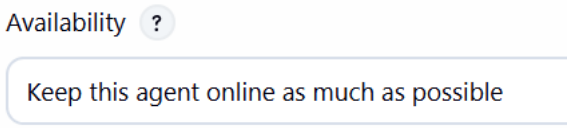


12. Created credentials as username with private key

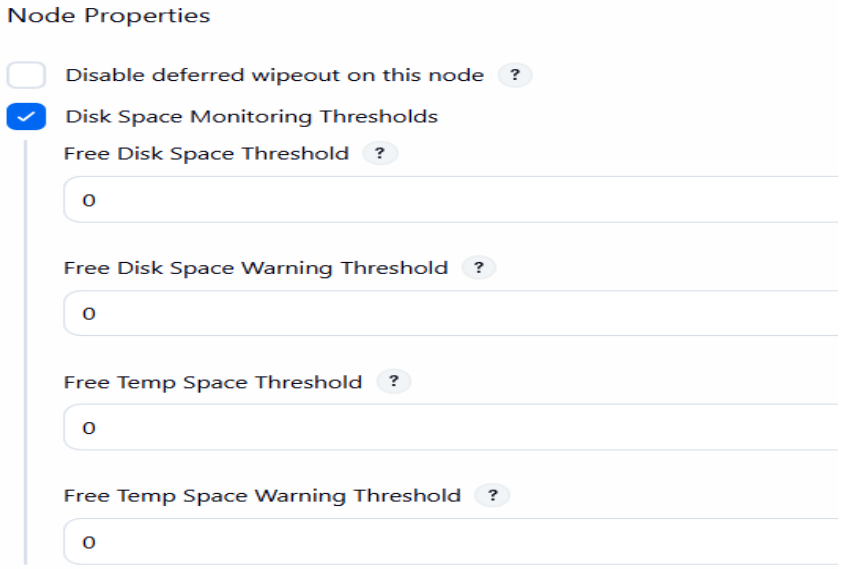


13. Selected launch method as “Launch agent via SSH” and provided private Ip and attached credentials and selected manually tested key verification strategy.

14. Selected availability as “Keep this agent online as much as possible”



15. Made all disk space monitoring thresholds as 0 in node properties and then clicked to apply and save



16. Now we have 1 slave node that we just created

S	Name ↓	Architecture	Clock Difference	Free Disk Space	Free Swap Space	Free Temp Space	Response Time
	Built-In Node	Linux (amd64)	In sync	5.58 GiB	0 B	5.58 GiB	0ms
	slave-1	Linux (amd64)	In sync	6.03 GiB	0 B	458.23 MiB	24ms
	Data obtained	**32 min**	**32 min**	**32 min**	**32 min**	**32 min**	**32 min**

17. Gave /mnt folder full permission

```
[root@ip-172-31-47-245 /]# chmod -R 777 /mnt/
```

18. Creating freestyle job on master

New Item

Enter an item name

assignment-9

Select an item type

Pipeline
Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.

Freestyle project
Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.

Multi-configuration project
Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.

19. Selected “Restrict where this project can run” as built in

Restrict where this project can be run ?

Label Expression ?

built-in

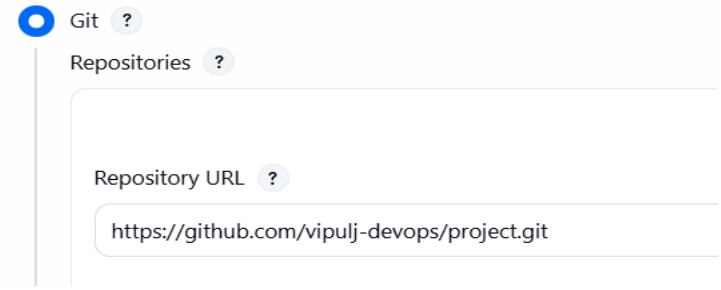
20. Added custom workspace

Use custom workspace ?

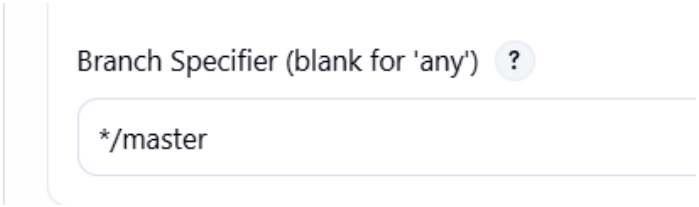
Directory

/mnt/project

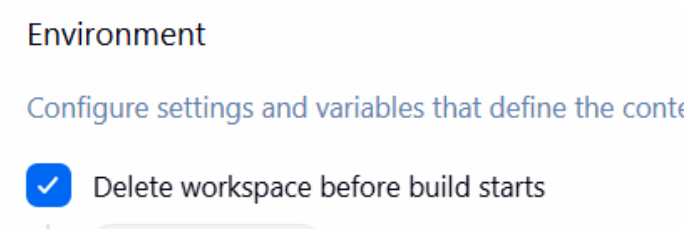
21. Put GitHub repository URL in GitHub URL section of Source Code Management



22. Select master branch in Branch Specifier



23. Check on 'Delete workspace before build starts' option. So, every time when we start new build the workspace will be cleared



24. Now in build step wrote first shell script to clean the maven repository

25. In build step selected Invoke top-level Maven targets configured Maven version and specified the goal in which mentioned two maven tasks (clean and install)

Invoke top-level Maven targets ?

Maven Version

apache-maven-3.9.13

Goals

clean
install

26. Wrote the below shell script in execute shell section and clicked on apply and save

See the list of available environment variables

```
#!/bin/bash

# Variables
SLAVE_IP="15.206.73.109"
KEY="/mnt/other/linuxkey.pem"
WAR_PATH="/mnt/projects/target/LoginWebApp.war"
TOMCAT_PATH="/mnt/servers/apache-tomcat-10.1.52/webapps"

echo "Starting deployment..."

# Step 1: Remove old app on slave
ssh -i $KEY ec2-user@$SLAVE_IP << EOF
rm -rf $TOMCAT_PATH/LoginWebApp*
EOF

# Step 2: Copy new WAR file
scp -i $KEY $WAR_PATH ec2-user@$SLAVE_IP:$TOMCAT_PATH/

# Step 3: (Optional but good) Restart Tomcat
ssh -i $KEY ec2-user@$SLAVE_IP << EOF
$TOMCAT_PATH/../bin/startup.sh
EOF

echo "Deployment completed successfully."
```

27. After clicking build now option the build ran successfully

#2 (Mar 17, 2026, 7:20:20 PM)

Started by user vipul

This run spent:

- 49 ms waiting;
- 15 sec build duration;
- 15 sec total from scheduled to completion.

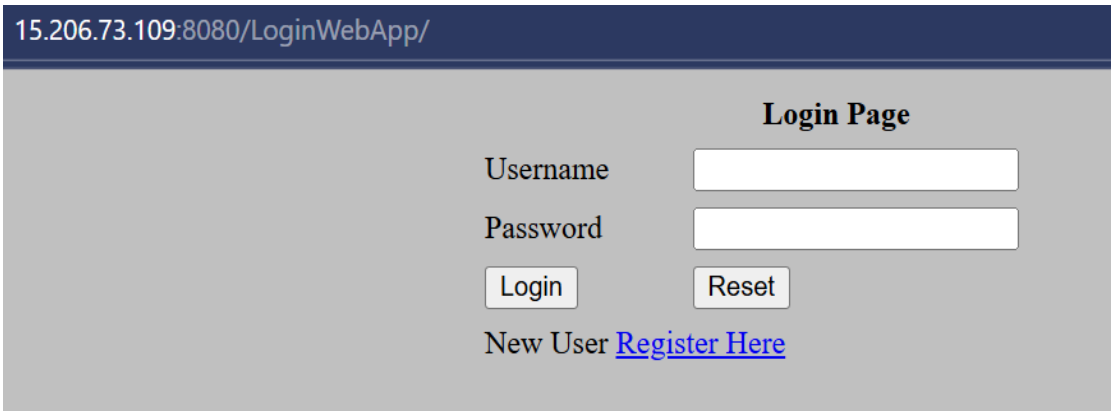
git

Revision: dad4cbb1657dd0bc83e677cdf2588836f9c329f2
Repository: https://github.com/vipulj-devops/project.git

- refs/remotes/origin/master

No changes.

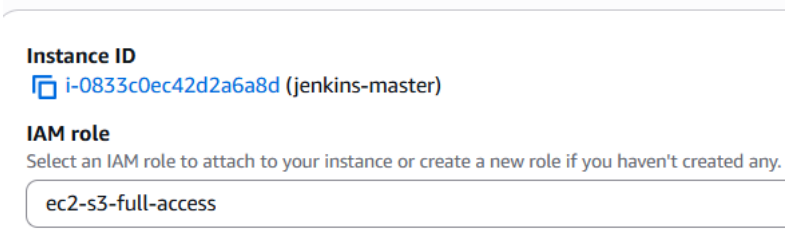
28. War file got hosted successfully on slave tomcat server

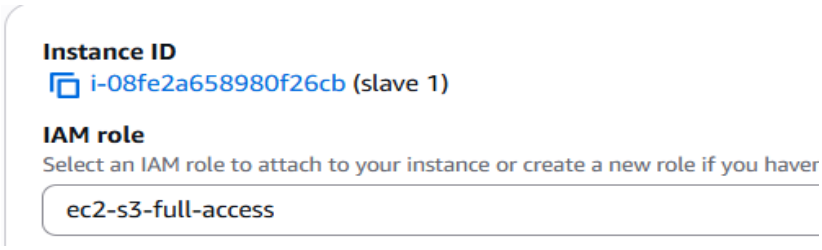


Task (b): upload the WAR file generated on the Jenkins master to an Amazon S3 bucket and deploy it on a Tomcat server running on a slave instance by downloading the artifact from S3**.**

Steps:

1. Attach ec2-s3-full-access role to master and slave instance.





2. Now keeping everything same in the job configuration, we just have to make few changes in the script.

```bash
#!/bin/bash
# Variables
SLAVE_IP="15.206.73.109"
KEY="/mnt/other/linuxkey.pem"
WAR_PATH="/mnt/projects/target/LoginWebApp.war"
BUCKET_NAME="vj-assignment-9"
TOMCAT_PATH="/mnt/servers/apache-tomcat-10.1.52/webapps"
WAR_NAME="LoginWebApp.war"

echo "Starting deployment with S3..."

# Step 1: Upload WAR to S3
echo "Uploading WAR to S3..."
aws s3 cp $WAR_PATH s3://$BUCKET_NAME/$WAR_NAME

# Step 2: Remove old app on slave
echo "Cleaning old deployment on slave..."
ssh -i $KEY ec2-user@$SLAVE_IP << EOF
rm -rf $TOMCAT_PATH/LoginWebApp*
EOF

# Step 3: Download WAR from S3 on slave
echo "Downloading WAR on slave from S3..."
ssh -i $KEY ec2-user@$SLAVE_IP << EOF
aws s3 cp s3://$BUCKET_NAME/$WAR_NAME $TOMCAT_PATH/
EOF

# Step 4: Restart Tomcat (recommended)
echo "Restarting Tomcat..."
ssh -i $KEY ec2-user@$SLAVE_IP << EOF
$TOMCAT_PATH/../bin/startup.sh
EOF

echo "Deployment via S3 completed successfully."
```

3. Job ran successfully

#4 (Mar 17, 2026, 8:27:30 PM)

Started by user vipul

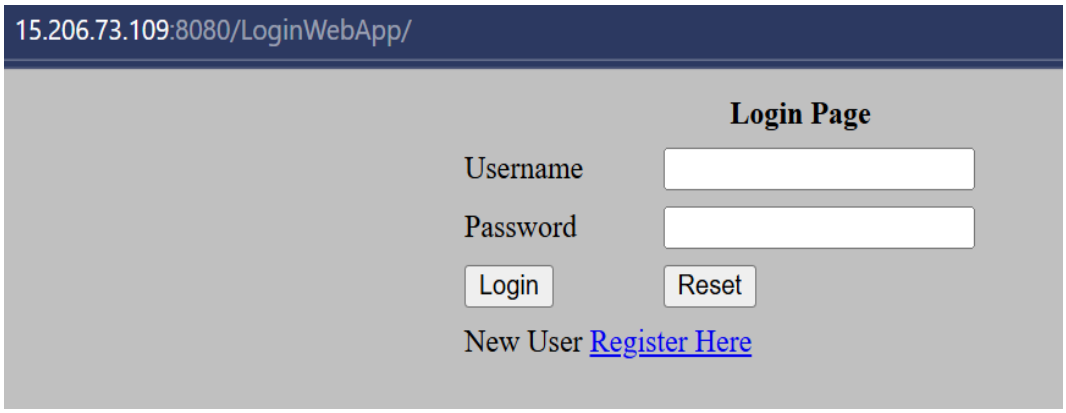
This run spent:

- 6 ms waiting;
- 16 sec build duration;
- 16 sec total from scheduled to completion.

git

Revision: dad4cbb1657dd0bc83e677cdf2588836f9c329f2
Repository: https://github.com/vipulj-devops/project.git

- refs/remotes/origin/master

No changes.

4. War got deployed successfully and website got hosted